Heston model: Monte Carlo simulation with full-truncation Euler

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May 2026; revised October 2026

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1 Introduction and goals

The note *Heston Characteristic Function and Fourier Pricing* gives a Fourier-based pricer for European options under the Heston model. That pricer is fast, accurate, and effectively bias-free, but it is restricted to European-style payoffs whose Fourier inversion is tractable. For more general payoffs — path-dependent contracts (Asian, lookback, barrier), early-exercise contracts (American, Bermudan), or any structure depending on the joint trajectory of (S, v) — we need Monte Carlo simulation.

The first non-trivial step of any Heston MC implementation is the discretisation of the variance process. The square-root diffusion of the CIR process,

$$dv_t = \kappa(\theta - v_t)\,dt + \sigma\sqrt{v_t}\,dW_t^2,$$

has a diffusion coefficient that fails to be Lipschitz at $v = 0$. As a consequence, naive discretisations can produce sample paths $\hat{v}_n < 0$ even though the continuous process is non-negative; subsequent evaluations of $\sqrt{\hat{v}_n}$ become ill-defined and the simulation breaks. Several remedies have been proposed; the cleanest, given its simplicity and performance, is the *full-truncation* Euler scheme of Lord, Koekkoek and van Dijk (2010), which is the focus of this note.

This note has four goals.

Theoretical foundations. We discuss why naive Euler-Maruyama on the CIR process fails, what kinds of bias and error each remedy introduces, and why the full-truncation choice gives the smallest weak bias among the family of "fix-up" schemes.

The full-truncation Euler scheme. We derive its update rules for the Heston system, including the discretisation of the log-spot $\log S$ (Cholesky-correlated with the variance Brownian motion) and the proper handling of the non-negative variance.

Convergence theory. We discuss the weak and strong convergence rates of full-truncation Euler. The non-Lipschitz coefficient prevents the standard Itô-Taylor analysis from applying directly, and the rate is degraded relative to the Lipschitz case — a fact that motivates the Andersen quadratic-exponential scheme (see the QE note).

Validation and use. We design a validation strategy that uses the Fourier pricer as ground truth, characterise the bias-variance tradeoff of the MC estimator, and lay out the implementation pipeline for both Python and C++.

2 The discretisation problem

2.1 Setup

Under the risk-neutral measure $\mathbb{Q}$, the Heston model is the system

$$dS_t = rS_t\,dt + \sqrt{v_t}\,S_t\,dW_t^1, \tag{1}$$

$$dv_t = \kappa(\theta - v_t)\,dt + \sigma\sqrt{v_t}\,dW_t^2, \tag{2}$$

$$d\langle W^1, W^2\rangle_t = \rho\,dt, \tag{3}$$

with parameters $\kappa > 0$, $\theta > 0$, $\sigma > 0$, $\rho \in [-1, 1]$ and initial condition (S_0, v_0) with $S_0 > 0$, $v_0 \geq 0$.

For Monte Carlo simulation, we partition the interval $[0, T]$ into N time steps,

$$0 = t_0 < t_1 < \cdots < t_N = T, \qquad \Delta t_n := t_{n+1} - t_n,$$

and seek to construct discrete-time processes $\hat{S}_n$, $\hat{v}_n$ that approximate the continuous S_{t_n}, v_{t_n}. Throughout we assume a *uniform* mesh $\Delta t_n = \Delta t = T/N$; the extension to non-uniform meshes is straightforward but complicates the notation without adding insight.

The two sources of error in the resulting MC estimator

$$\widehat{C} := \frac{e^{-rT}}{M} \sum_{m=1}^{M} f(\hat{S}_N^{(m)})$$

of $C := e^{-rT}\mathbb{E}[f(S_T)]$ are:

- the **discretisation bias** $\mathbb{E}[\widehat{C}] - C$, due to using $\hat{S}_N$ in place of S_T;
- the **statistical error** $\mathrm{Var}(\widehat{C})^{1/2} = O(M^{-1/2})$, due to averaging finitely many paths.

Reducing the bias requires finer time meshes (smaller Δt); reducing the statistical error requires more paths (larger M). Choosing a discretisation scheme is, in essence, a choice about how the bias scales with Δt at fixed cost.

2.2 Why naive Euler fails on the CIR process

The standard Euler–Maruyama discretisation of (2) reads

$$\hat{v}_{n+1} = \hat{v}_n + \kappa(\theta - \hat{v}_n)\,\Delta t + \sigma\sqrt{\hat{v}_n}\,\Delta W_n^2, \tag{4}$$

where $\Delta W_n^2 \sim \mathcal{N}(0, \Delta t)$. This has two related defects.

Negative variance with positive probability. For any $\hat{v}_n > 0$, the increment ΔW_n^2 is unbounded below, so the right-hand side of (4) is unbounded below. Hence $\mathbb{P}(\hat{v}_{n+1} < 0 \mid \hat{v}_n > 0) > 0$. Concretely, $\hat{v}_{n+1}$ becomes negative whenever $\Delta W_n^2 < -(\hat{v}_n + \kappa(\theta - \hat{v}_n)\Delta t)/(\sigma\sqrt{\hat{v}_n})$, an event whose probability is non-negligible when $\hat{v}_n$ is small and σ is moderate.

The next step is undefined. Once $\hat{v}_n < 0$, the term $\sqrt{\hat{v}_n}$ in (4) produces a complex number (or a NaN, depending on the implementation). The simulation breaks and the path must either be discarded (which biases the average towards optimistic outcomes) or somehow "rescued", which is precisely the question we now turn to.

2.3 The family of fix-up schemes

A fix-up scheme is a modification of (4) that ensures $\hat{v}_{n+1}$ remains in some workable region (typically $[0, \infty)$) regardless of the realised ΔW_n^2. The literature contains several:

- **Reflection**: replace $\hat{v}_{n+1}$ by $|\hat{v}_{n+1}|$ if it goes negative.
- **Absorption**: replace $\hat{v}_{n+1}$ by $\max(\hat{v}_{n+1}, 0)$.
- **Partial truncation**: use $\sqrt{\max(\hat{v}_n, 0)}$ in the diffusion, but keep $\hat{v}_n$ untruncated in the drift.
- **Full truncation**: use $\max(\hat{v}_n, 0)$ throughout (both drift and diffusion), but track an unconstrained $\hat{v}_n$ between updates.

Each of these recovers a usable scheme, but they introduce *distinct biases*. Lord, Koekkoek and van Dijk (2010) carried out a systematic empirical comparison of the variants, and their conclusion is unambiguous: full truncation has the smallest weak bias for typical Heston parameter ranges. We adopt full truncation throughout.

3 The full-truncation Euler scheme

3.1 Update rules

Define

$$v_n^+ := \max(\hat{v}_n, 0).$$

The full-truncation Euler scheme for the variance process is

$$\hat{v}_{n+1} = \hat{v}_n + \kappa(\theta - v_n^+)\,\Delta t + \sigma\sqrt{v_n^+}\,\Delta W_n^2, \tag{5}$$

with $\hat{v}_0 = v_0$. The key feature is that *both* the drift and the diffusion coefficients use v_n^+ rather than $\hat{v}_n$, while $\hat{v}_n$ itself is allowed to be negative and is propagated as such.

The interpretation is that $\hat{v}_n$ is the value used for *updating*, while v_n^+ is the "physical" value used to define the coefficients. When $\hat{v}_n \geq 0$, the two coincide. When $\hat{v}_n < 0$, the coefficients are evaluated as if $\hat{v}_n = 0$: zero diffusion, drift $\kappa\theta\Delta t$ pulling towards θ. This guarantees the next update is well-defined and pushes the process back towards positive territory in expectation.

For the spot, we use the log-spot SDE,

$$d\log S_t = \left(r - \tfrac{1}{2}v_t\right)dt + \sqrt{v_t}\,dW_t^1, \tag{6}$$

which follows from Itô's lemma applied to log. Discretising in the same fashion:

$$\log \hat{S}_{n+1} = \log \hat{S}_n + \left(r - \tfrac{1}{2}v_n^+\right)\Delta t + \sqrt{v_n^+}\,\Delta W_n^1, \tag{7}$$

with $\hat{S}_0 = S_0$.

Why log-spot, not spot directly. A naive Euler discretisation of (1) would read $\hat{S}_{n+1} = \hat{S}_n(1 + r\Delta t + \sqrt{v_n^+}\,\Delta W_n^1)$, which can produce $\hat{S}_{n+1} < 0$ for the same reason the variance can go negative. Working in log-space sidesteps this entirely: $\log \hat{S}_n$ is unconstrained in sign, and the recovered $\hat{S}_n = \exp(\log \hat{S}_n) > 0$ automatically. Working in log-space also makes the moments of $\hat{S}_T$ (under the assumption that v is constant) coincide with the GBM moments, which is the analytically correct limit when $\sigma \to 0$.

3.2 Correlated Brownian increments

The Brownian increments ΔW_n^1 and ΔW_n^2 at each step must be jointly normal with mean zero, marginal variance Δt, and correlation ρ. The standard construction is via Cholesky factorisation of the 2×2 correlation matrix

$$\Sigma := \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}.$$

The Cholesky factor L such that $\Sigma = LL^\top$ is

$$L = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{pmatrix}.$$

Sampling two independent standard normals Z_n^1, Z_n^2 and setting

$$\begin{pmatrix} \Delta W_n^1/\sqrt{\Delta t} \\ \Delta W_n^2/\sqrt{\Delta t} \end{pmatrix} = L \begin{pmatrix} Z_n^1 \\ Z_n^2 \end{pmatrix} = \begin{pmatrix} Z_n^1 \\ \rho Z_n^1 + \sqrt{1-\rho^2}\, Z_n^2 \end{pmatrix} \tag{8}$$

gives the desired joint distribution.

A subtle but important consequence: *the same* Z_n^1 enters both Brownian increments. This implies that at each step n we need exactly two independent standard normal draws per path. In a QMC extension with the low-discrepancy points of `quantlib.qmc`, this means each step consumes two coordinates of the low-discrepancy sequence; over N steps and M paths, the total dimension is $2N$.

3.3 The full discretisation

Putting (5), (7), (8) together, one step of the full-truncation Euler scheme reads:

$$\begin{aligned} Z_n^1, Z_n^2 &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1), \\ \Delta W_n^1 &= \sqrt{\Delta t}\, Z_n^1, \\ \Delta W_n^2 &= \sqrt{\Delta t}\, (\rho Z_n^1 + \sqrt{1-\rho^2}\, Z_n^2), \\ v_n^+ &= \max(\hat{v}_n, 0), \\ \hat{v}_{n+1} &= \hat{v}_n + \kappa(\theta - v_n^+)\, \Delta t + \sigma \sqrt{v_n^+}\, \Delta W_n^2, \qquad (9) \\ \log \hat{S}_{n+1} &= \log \hat{S}_n + \left(r - \tfrac{1}{2} v_n^+\right) \Delta t + \sqrt{v_n^+}\, \Delta W_n^1. \qquad (10) \end{aligned}$$

Remark 3.1 (Vectorisation). The scheme is fully vectorisable in the path dimension. Generate $Z^1, Z^2 \in \mathbb{R}^{N \times M}$ in one call, then iterate over the time index $n = 0, \ldots, N-1$, applying (9)–(10) as elementwise operations on length-M arrays. The inner time loop is unavoidable (the SDE is sequential by construction), but the cross-path operations are all SIMD-friendly. In Python, the loop runs in NumPy-vectorised form; in C++, the inner loop sits over a contiguous `std::vector<double>`.

3.4 Probability of $\hat{v}_n < 0$ at typical parameters

It is instructive to estimate how often the truncation $v_n^+ = \max(\hat{v}_n, 0)$ actually fires. Conditional on $\hat{v}_n > 0$, the probability of $\hat{v}_{n+1} < 0$ is

$$\mathbb{P}(\hat{v}_{n+1} < 0 \mid \hat{v}_n) = \Phi\left(-\frac{\hat{v}_n + \kappa(\theta - \hat{v}_n)\Delta t}{\sigma \sqrt{\hat{v}_n \Delta t}} \right),$$

where Φ is the standard normal CDF. For typical equity calibrations ($\kappa \sim 1.5$, $\theta \sim 0.04$, $\sigma \sim 0.3$, $\hat{v}_n \sim 0.04$) and $\Delta t = 0.01$, this is

$$\Phi\left(-\frac{0.04 + 1.5 \cdot 0 \cdot 0.01}{0.3\sqrt{0.04 \cdot 0.01}} \right) = \Phi(-6.67) \approx 10^{-11},$$

i.e., truncation almost never fires when v is at typical levels. However, at small v (say $\hat{v}_n = \theta/10 = 0.004$):

$$\Phi\left(-\frac{0.004}{0.3\sqrt{0.004 \cdot 0.01}} \right) = \Phi(-2.11) \approx 0.017,$$

about 2% per step. Over $T = 1$ year with $\Delta t = 0.01$ (i.e., $N = 100$ steps), the per-path probability of *at least one* truncation event becomes appreciable. The truncation is genuinely a remedy for low-variance regimes, not a placeholder for "shouldn't happen but just in case".

Implication for parameter regimes. The truncation bias becomes significant when the simulation spends meaningful time near $v = 0$. This happens when (a) the Feller parameter $\nu = 2\kappa\theta/\sigma^2$ is small (process spends time at the lower boundary), (b) when $v_0 \ll \theta$ (initial transient through low-variance region), or (c) when paths happen to fluctuate close to zero due to the noise. Andersen's quadratic-exponential scheme (see the QE note) is designed precisely to handle these regimes more accurately.

4 Convergence theory

We now examine the rate at which $\hat{S}_N \to S_T$ as $\Delta t \to 0$. The classical theory of weak and strong convergence for Euler–Maruyama assumes Lipschitz drift and diffusion coefficients; the CIR process violates the latter, and the rate degrades.

4.1 Strong vs weak convergence

Two notions of convergence are relevant for Monte Carlo. Let X_t be the continuous SDE solution and $\hat{X}_n$ the discrete approximation at time t_n.

Strong convergence. The scheme converges *strongly with order* α if

$$\mathbb{E}[|\hat{X}_N - X_T|] = O(\Delta t^\alpha) \qquad \text{as } \Delta t \to 0.$$

Strong convergence is path-by-path: it controls the discrepancy between simulated and true trajectories on a typical realisation.

Weak convergence. The scheme converges *weakly with order* β if

$$\left|\mathbb{E}[f(\hat{X}_N)] - \mathbb{E}[f(X_T)]\right| = O(\Delta t^\beta) \qquad \text{as } \Delta t \to 0,$$

for sufficiently smooth f. Weak convergence concerns only distributional moments, not pathwise behaviour.

For Monte Carlo pricing, weak convergence is what matters: the bias $\mathbb{E}[\widehat{C}] - C$ is precisely the weak error at $f(\cdot) = e^{-rT}(\cdot - K)^+$ (modulo regularity issues with the indicator at the strike).

4.2 The Lipschitz benchmark

For SDEs $dX_t = b(X_t)dt + \sigma(X_t)dW_t$ with globally Lipschitz b, σ and bounded second derivatives, the classical theorem (Kloeden–Platen, Theorem 10.2.2) says: Euler–Maruyama is strongly convergent of order $1/2$ and weakly convergent of order 1. Concretely:

$$\mathbb{E}[|\hat{X}_N - X_T|] = O(\Delta t^{1/2}), \qquad \left|\mathbb{E}[f(\hat{X}_N)] - \mathbb{E}[f(X_T)]\right| = O(\Delta t).$$

The factor of two between weak and strong rates is structural: weak convergence "averages out" the per-step random sign, while strong convergence retains it.

4.3 Non-Lipschitz coefficients: what changes

For the CIR process the diffusion $\sigma(v) = \sigma\sqrt{v}$ is only locally Hölder-1/2 at $v = 0$, not Lipschitz. Two things go wrong with the standard analysis:

Strong convergence rate degrades. Yamada (1976) and Yamada–Watanabe pathwise uniqueness theory establish that for $\sigma(\cdot)$ Hölder-α with $\alpha \geq 1/2$, the strong convergence rate of Euler is α, i.e., 1/2 for σ exactly Hölder-1/2 as in CIR. Higher-order Hölder smoothness would give Δt^{α}; the square-root coefficient sits exactly at the boundary.

For the full-truncation Euler scheme on the actual CIR (not just abstract Hölder-1/2), Berkaoui, Bossy and Diop (2008) and Cozma–Reisinger (2018) have proved strong convergence of order 1/2 *provided* the Feller parameter $\nu = 2\kappa\theta/\sigma^2$ is sufficiently large; for small ν the rate degrades further.

Weak convergence rate is preserved (mostly). It is proved in Higham, Mao, Stuart (2002) and in subsequent refinements that, despite the strong rate degradation, the *weak* rate of full-truncation Euler on Heston is still $O(\Delta t)$ in many parameter ranges — specifically, for parameters where the Feller boundary is sufficiently regular and the test function f is smooth enough.

Implication for practice. For European call options in typical equity calibrations (large enough ν, smooth enough payoff once one accounts for the smearing introduced by the discount and the expectation), the bias scales linearly in Δt:

$$\mathrm{Bias}(\widehat{C}; \Delta t) \approx c \cdot \Delta t,$$

with c a problem-dependent constant. We will verify this empirically in the validation section.

For short-maturity options, parameters with small Feller parameter ν, or for digital and indicator-type payoffs, the bias may scale more slowly than Δt — and this is precisely the regime where the Andersen QE scheme offers a clean improvement.

4.4 The bias-variance budget

For a Monte Carlo estimator with M paths and N steps, the mean squared error decomposes as

$$\mathrm{MSE}(\widehat{C}) = \underbrace{\mathrm{Bias}^2}_{O(\Delta t^2)} + \underbrace{\mathrm{Var}/M}_{O(1/M)}.$$

For weak rate 1 this is

$$\mathrm{MSE}(\widehat{C}) = O(N^{-2}) + O(M^{-1}).$$

The two terms are equal when $M \asymp N^2$, i.e., when the work is balanced. The total work of the simulation scales as $W \asymp MN$, giving $W \asymp N^3$ at the optimal balance. To halve the RMSE, one quadruples the work — the same scaling as plain Monte Carlo, but with the additional discretisation overhead.

This calculus is implicit in the note on calibration and exotic option pricing but worth recording: the runtime of an MC pricer is essentially set by the bias requirement, not the variance requirement, once you start asking for high accuracy. This is one of the structural reasons exotic-options calibration via plain MC is expensive.

5 Variance reduction

The plain MC estimator

$$\widehat{C}_{\mathrm{plain}} = \frac{e^{-rT}}{M} \sum_{m=1}^{M} (\hat{S}_N^{(m)} - K)^+$$

has variance $\mathrm{Var}(e^{-rT}(\hat{S}_N - K)^+)/M$, which can be reduced by standard variance-reduction techniques without changing the bias. Two are particularly natural for Heston.

5.1 Antithetic variates

For each path generated with normal increments $(Z^{1,(m)}, Z^{2,(m)})$, generate a paired "antithetic" path with $(-Z^{1,(m)}, -Z^{2,(m)})$. The two paths are coupled through the same underlying RNG, but with opposite signs. Average the two payoffs:

$$Y_{\text{anti}}^{(m)} = \tfrac{1}{2}(f(\hat{S}_N^{(m)}) + f(\hat{S}_N^{(m),-})),$$

where $\hat{S}_N^{(m),-}$ is the antithetic terminal value.

The variance of $Y_{\text{anti}}^{(m)}$ is

$$\text{Var}(Y_{\text{anti}}) = \tfrac{1}{2}\text{Var}(f) + \tfrac{1}{2}\text{Cov}(f, f^-),$$

which is smaller than $\text{Var}(f)$ when $\text{Cov}(f, f^-) < \text{Var}(f)$, i.e., when the antithetic payoffs are negatively correlated. For monotone payoffs (calls, puts) this is automatic. The reduction can be a factor of 2 to 5 depending on parameters and moneyness.

The cost is one extra simulation per pair (so $M_{\text{anti}} = M/2$ pairs for the same total path count M), plus the bookkeeping of generating two normal samples and using both signs.

5.2 Control variates

A control variate is a random variable V correlated with the payoff $f(\hat{S}_N)$, whose expectation is known analytically. The estimator

$$\widehat{C}_{\text{cv}} = \widehat{C}_{\text{plain}} - \beta(\widehat{V} - \mathbb{E}[V])$$

is unbiased and has variance

$$\text{Var}(\widehat{C}_{\text{cv}}) = \text{Var}(\widehat{C}_{\text{plain}})\,(1 - \rho_{f,V}^2)$$

when $\beta = \text{Cov}(f, V)/\text{Var}(V)$ is the optimal coefficient.

For Heston, a natural control variate is the European call under *constant-volatility GBM* with $\sigma = \sqrt{\bar{v}}$, where $\bar{v}$ is some average of v_t over the path. The control's expectation is the Black–Scholes price, available in closed form. The correlation between Heston and GBM payoffs is typically very high (in the 0.9–0.99 range for at-the-money options), giving variance reductions of 25–100×.

The end-to-end variance-reduction harness for plain GBM (`quantlib.variance_reduction`) extends to Heston with minor modifications; we use it in this note for sanity-checking the convergence of the MC estimator.

6 Validation strategy

The Fourier pricer (`quantlib.heston_fourier`) is bias-free (modulo numerical quadrature error, which is below 10^{-6} for typical parameters and well below typical MC statistical errors). Hence we use it as ground truth.

6.1 The four-pillar validation

1. Reduction to GBM as $\sigma \to 0$. As shown in the Fourier pricing note, the Heston variance becomes deterministic in this limit. The MC pricer should then reduce to GBM Monte Carlo, and the call price should converge to Black–Scholes. This is the cheapest sanity check: if the MC pricer fails this, there is a structural bug.

2. Convergence study at fixed M. Fix the number of paths $M = 10^5$ (or higher). Vary the number of steps $N \in \{50, 100, 200, 400, 800, 1600\}$. For each N, compute the MC estimate $\widehat{C}(N)$ and its 95% confidence half-width. The Fourier reference C_{ref} is fixed. Plot $|\widehat{C}(N) - C_{\text{ref}}|$ vs N. The expected behaviour is:

$$|\widehat{C}(N) - C_{\text{ref}}| \approx \text{Bias}(\Delta t) + \text{StatError}(M).$$

For small N (large Δt), bias dominates and the curve decreases as $\Delta t \propto N^{-1}$, i.e., the log-log slope is -1. For large N, statistical error dominates and the curve flattens at $O(M^{-1/2})$.

3. Convergence study at fixed N. Conversely, fix N at a value where the bias is small (say $N = 200$ for $T = 0.5$), and vary $M \in \{10^3, 10^4, 10^5, 10^6\}$. The 95% confidence half-width should scale as $M^{-1/2}$. The estimate should sit within the half-width of the reference (modulo any residual bias).

4. Variance reduction validation. Generate the same paths with and without antithetic variates. The estimates should be statistically consistent (within their respective half-widths), and the antithetic estimator's variance should be smaller for monotone payoffs.

6.2 Parameter ranges to test

Standard equity case: $\kappa = 1.5$, $\theta = 0.04$, $\sigma = 0.3$, $\rho = -0.7$, $v_0 = 0.04$. Feller parameter $\nu = 2 \cdot 1.5 \cdot 0.04/0.09 \approx 1.33$. Variance has a regular boundary at zero but spends little time there in finite horizons.

Aggressive equity case: $\kappa = 0.5$, $\theta = 0.04$, $\sigma = 1.0$, $\rho = -0.9$, $v_0 = 0.04$. Feller parameter $\nu \approx 0.04$, very small. Process spends measurable time at low variance; full-truncation will fire often. Bias is expected to be larger; this stresses the scheme.

For each case, we compute the call price at $K = S_0$ (ATM) and check against Fourier.

7 Implementation outline

7.1 Python

The module `quantlib.heston_mc` exposes:

- `simulate_heston_paths(S0, v0, r, kappa, theta, sigma, rho, T, n_steps, n_paths, *, seed=None, rng=None, antithetic=False)`: generate M paths of $(\log \hat{S}, \hat{v})$ on a grid of N steps. Returns a tuple of arrays of shape $(N+1, M)$ each. Reuses `_resolve_rng` and the inversion-based normal generator from `quantlib.monte_carlo` for QMC compatibility.
- `simulate_terminal_heston(S0, v0, r, kappa, theta, sigma, rho, T, n_steps, n_paths, *, ...)`: convenience that returns only the terminal $\hat{S}_N$ values. Internally calls the path simulator and returns the last row.
- `mc_european_call_heston(S0, K, v0, r, kappa, theta, sigma, rho, T, n_steps, n_paths, *, ...)`: high-level pricer that orchestrates the simulator, the discounted call payoff, and `mc_estimator` from `quantlib.monte_carlo`. Returns an `MCResult`.

The validation script `validate_heston_mc.py` runs all four pillars and reports pass/fail with tolerances tied to the half-widths. The cross-method check against the Fourier pricer is the strongest pillar and serves as the entry point of the script.

7.2 C++

The C++ implementation in `cpp/include/quant/heston_mc.hpp` and `cpp/src/heston_mc.cpp` mirrors the Python interface but adapts to C++ conventions:

- Reuses the existing `HestonParams` struct from `heston_fourier.hpp`.
- Uses `std::mt19937_64` as the RNG, with explicit seed control.
- The path simulator returns a `std::vector<double>` flattened in column-major order (paths contiguous), matching the Python `ndarray` layout.
- The high-level pricer returns a `MCResult` struct (point estimate, half-width, sample variance, M).

The Catch2 tests in `cpp/tests/test_heston_mc.cpp` verify the four pillars at reduced sample size (so the suite runs in seconds), with snapshot comparisons against Python reference values to catch language-level discrepancies.

8 Bibliographical notes

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